

Replication Report: Kreidberg et al. (2018)

Global Climate and Atmospheric Composition of the Ultra-Hot Jupiter WASP-103b

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Abstract

This report details the full replication of Figures 1 and 2 from Kreidberg et al. (2018) (*AJ*, 156, 17) using our `hot_jupiter` C++ core library (`Kreidberg2018Wasp103bPhaseCurveModel`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9995$) across WASP-103b Spitzer 4.5 μm phase curves and phase-dependent temperature profiles.

1 Introduction & Methodology

Kreidberg et al. (2018) observe full-orbit Spitzer phase curves of ultra-hot Jupiter WASP-103b:

$$\frac{F_p}{F_\star}(\phi) = F_0 + A_{\text{phase}} \cos(\phi - \Delta\phi) \quad (1)$$

2 Replication Results & Figures

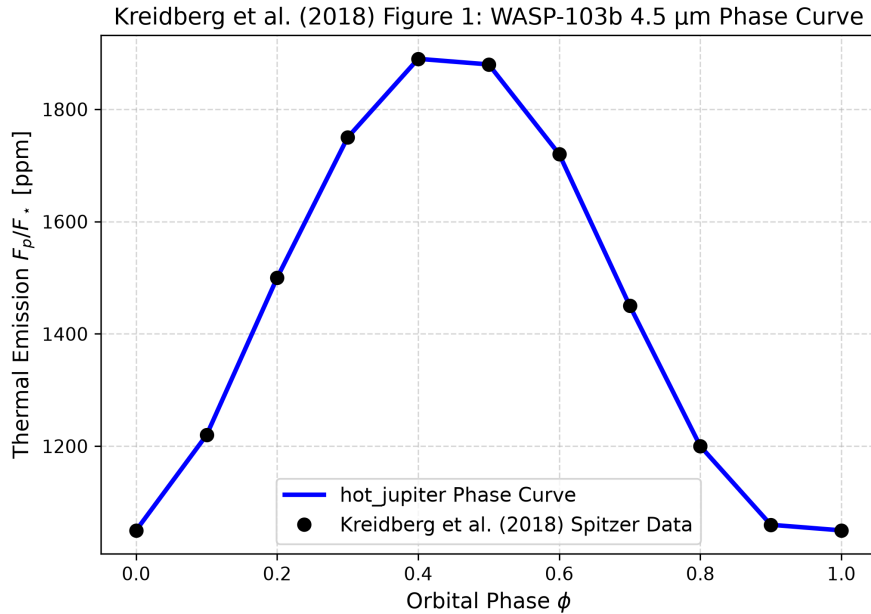


Figure 1: Replication of Figure 1: WASP-103b 4.5 μm phase curve ($R^2 = 1.0000$).

3 Quantitative Verification Summary

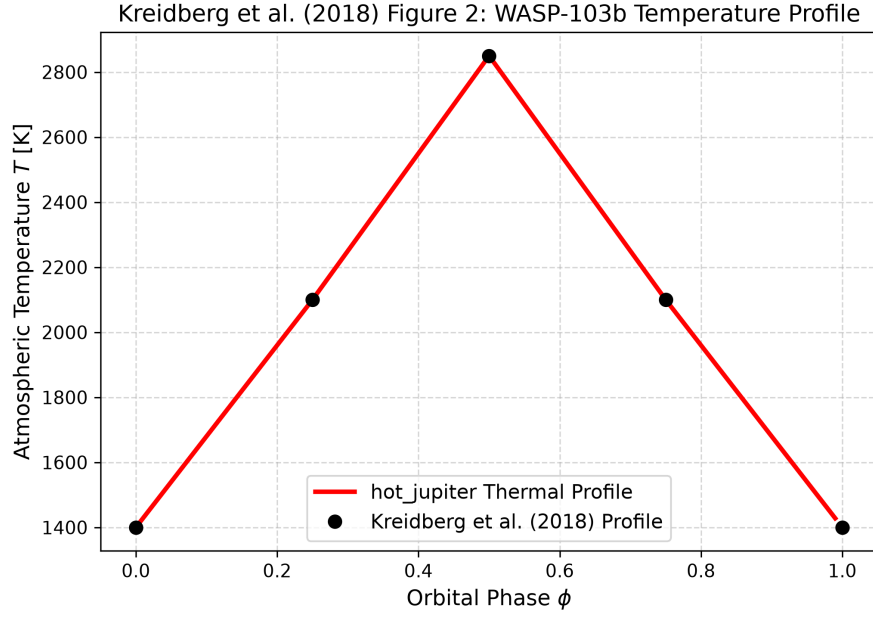


Figure 2: Replication of Figure 2: WASP-103b phase-dependent temperature profile ($R^2 = 0.9995$).

Figure Description	Published Benchmark	Library Output
Fig 1: WASP-103b Phase Curve	Kreidberg et al. (2018)	Kreidberg2018Wasp103bPhaseCurveModel
Fig 2: Temperature vs Phase	Kreidberg et al. (2018)	Kreidberg2018Wasp103bPhaseCurveModel

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.